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THESIS EVALUATION AND SUBMISSION APPROVAL

To: The Internship and Thesis Jury 2025–2026,
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We, **MSc. Vũ Hà My** (External Supervisor) and **Dr. Nghiêm Thị Phương** (Internal Supervisor), certify that the graduation thesis entitled:

"Intelligent Weather-Aware Travel Assistant for Vietnam"

conducted by student **Đặng Đình Hòa** (Student ID: **23BI14169**), has completed all research and implementation requirements, and is qualified to be submitted and presented before the Graduation Jury of the academic year 2025–2026.

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List of Abbreviations

AI Artificial Intelligence

API Application Programming Interface

GIS Geographic Information System

JSON JavaScript Object Notation

LLM Large Language Model

MAE Mean Absolute Error

NLP Natural Language Processing

REST Representational State Transfer

RMSE Root Mean Square Error

UI User Interface

UX User Experience

WMO World Meteorological Organization

Abstract

Vietnam possesses a long and narrow geography characterized by complex tropical monsoon climates and frequent severe weather phenomena such as typhoons, torrential rains, dense fog, and thunderstorms. These meteorological factors pose critical travel safety risks, particularly for motorcyclists, who constitute the predominant proportion of commuters in the country.

This graduation project presents the research, design, and implementation of the "Smart Vietnam Weather Information and Safe Travel Route Planning System with AI Grounding" (VietForecast). The system addresses core technical challenges through the following key contributions:

1. Meteorological Ingestion and Spatial Normalization: Automated real-time harvesting across 63 Vietnamese provinces, persisted within a PostGIS spatial database with continuous validation of data authenticity, integrity, and temporal freshness.
2. Multi-Dimensional Weather Risk Evaluation Engine: A config-driven matrix evaluating hazards across precipitation intensity, wind gusts, surface visibility, extreme temperatures, and road surface conditions, customized by vehicle profile (motorcycles and automobiles).
3. Multi-Alternative Weather-Aware Route Optimization: Concurrent calculation of Fastest, Shortest, and Weather-Safer routes for inter-provincial corridors spanning up to 2,000 km.
4. Grounded AI Travel Assistant: Integration with Google Gemini LLM via function calling over internal tool declarations, ensuring responses are strictly grounded with verifiable spatial citations without hallucinations.
5. Containerized Architecture and Quota Governance: Segmented Docker networking, atomic Redis quota counters aligned with Asia/Ho Chi Minh midnight resets, and integrated digital payments.

Keywords: Weather-Aware Routing, Travel Risk Assessment, PostGIS, LLM Grounding, Fastify, MapLibre GL, Vietnam Meteorological Systems.

Chapter 1

Introduction and Problem Statement

1.1 Background and Motivation

Vietnam features a distinctive elongated geography spanning over 1,650 km from North to South, characterized by diverse topography ranging from rugged northern highlands and central mountain passes to expansive river deltas and an extensive coastline exceeding 3,260 km. Consequently, the country exhibits stark meteorological divergence across its climate sub-regions: subtropical humid conditions with cold winters in the North, intense tropical monsoons accompanied by severe typhoons in the Central region, and distinct wet-and-dry tropical savanna seasons in the South.

Extreme meteorological phenomena—including sudden cloudbursts, severe localized flash floods, high-velocity squall winds, dense mountain fog, and persistent urban inundation—occur with high frequency throughout the year. Adverse weather substantially increases traffic collision frequency, degrades tire-to-surface traction, and reduces driver visibility [1]. In Vietnam, this vulnerability is exceptionally acute due to the unique modal split of the transportation sector: over 72 million registered motorcycles constitute upwards of 85% of all active vehicular trips. Unlike occupants of enclosed passenger automobiles, two-wheeler riders remain directly exposed to physical weather elements. Heavy precipitation rapidly degrades tire-to-surface traction, strong lateral wind gusts destabilize balance across elevated bridges and coastal causeways, and fog reduces forward sight distance along hazardous mountain passes such as Pha Din, Hai Van, and O Quy Ho.

Despite the profound impact of weather conditions on personal safety and travel reliability, contemporary travelers lack unified digital tools that synthesize continuous spatial meteorology directly into path computation. Bridging the gap between empirical weather observations, geospatial algorithms, and intelligent decision support forms the core motivation of this thesis.

1.2 Existing Solutions and Their Limitations

Modern travelers predominantly rely on two disconnected categories of software applications: commercial geographic navigation platforms and standalone meteorological forecasting services.

1.2.1 Commercial Navigation Platforms

Leading turn-by-turn navigation applications, such as Google Maps, Apple Maps, and open-source OSRM implementations [2], utilize advanced graph contraction and routing algorithms [3]. However, their cost functions optimize almost exclusively for physical distance, free-flow travel duration, and historical or crowdsourced traffic congestion. These systems treat the atmosphere as a neutral medium. Meteorological hazards are either ignored entirely or presented as disconnected informational overlays without penalizing high-risk road corridors or providing proactive safety-aware alternatives.

1.2.2 Standalone Weather Forecasting Applications

Conversely, prominent meteorological applications (such as AccuWeather, Windy, and localized hydrometeorological portals) deliver detailed point-based forecasts and radar animations adhering to standard meteorological measurement guidelines [4]. Nonetheless, these tools operate in spatial isolation from user navigation. A commuter cannot assess the aggregate weather risk across a 300 km inter-provincial journey traversing multiple climate zones, nor can they determine whether departing 45 minutes later would allow them to bypass a convective storm cell.

1.2.3 Generic Generative Artificial Intelligence Tools

With the advent of Large Language Models (LLMs) [5, 6], users frequently inquire about travel advice through conversational interfaces. However, unaugmented models suffer from temporal cutoff constraints, lack access to real-time spatial databases, and frequently exhibit hallucinations—generating plausible yet factually incorrect weather claims. Without grounded retrieval and strict function calling protocols [7], conversational models cannot be safely relied upon for mission-critical travel decisions.

1.2.4 Comparative Analysis

Table 1.1 provides a structured comparison between existing platforms and the VietForecast system developed in this graduation project.

Table 1.1: Feature comparison between existing solutions and VietForecast.

Feature Dimension	Google Maps	Windy / AccuWeather	Generic LLMs	VietForecast (Proposed)
Turn-by-turn routing	Yes	No	No	Yes
Weather hazard path evaluation	No	No	No	Yes
Vehicle-specific risk weighting	No	No	No	Yes
Spatial corridor interpolation	No	Partial	No	Yes
Real-time verified observations	No	Yes	No	Yes
Grounded AI conversation	No	No	Partial	Yes
Open-source GIS reproducibility	No	No	No	Yes

1.3 Problem Statement and Research Questions

1.3.1 Problem Statement

Given an origin point O, a destination point D, a scheduled departure timestamp t_0, and a specific vehicle profile $V \in \{\text{Motorcycle}, \text{Automobile}\}$, the problem is to compute a set of alternative navigation paths $\mathcal{P} = \{P_{\text{fastest}}, P_{\text{shortest}}, P_{\text{safer}}\}$ such that each path is evaluated against real-time spatio-temporal weather hazard fields. Specifically, the weather-safer path P_{safer} must systematically minimize cumulative risk exposure while bounding the acceptable travel time elongation relative to P_{fastest}. Furthermore, the system must enable users to interactively query journey risks through a conversational assistant whose answers are strictly grounded in verifiable observational data.

1.3.2 Research Questions

To address this problem, this research investigates three primary scientific and engineering questions:

RQ1: *Spatial Data Freshness and Normalization:* How can heterogeneous meteorological streams across all 63 provinces in Vietnam be continuously ingested, validated, and normalized into a PostGIS spatial database [8] with sub-second retrieval latency?

RQ2: *Multi-Criteria Risk Quantification:* How can multi-modal physical parameters (precipitation rate, gust speed, horizontal visibility, surface wetness, and ambient temperature) be synthesized into an objective, vehicle-differentiated route hazard metric?

RQ3: *Grounded Conversational Assistance:* How can generative LLM architectures be constrained via deterministic tool calling to eliminate hallucinations and generate verifiable, context-aware travel advisories?

1.4 Project Objectives and Contributions

1.4.1 Specific Objectives

The concrete objectives of this graduation project comprise:

- Design and implement an automated meteorological pipeline that harvests, validates, and indexes real-time observational data across all 63 provinces of Vietnam.
- Develop an extensible, config-driven risk assessment engine that quantifies travel hazard scores tailored distinctly for two-wheelers and passenger cars.
- Integrate routing algorithms with geospatial weather surfaces to simultaneously generate Fastest, Shortest, and Weather-Safer routes for trips up to 2,000 km.
- Construct a conversational travel assistant powered by Gemini 1.5 with robust function calling against internal geospatial APIs.

– Deliver a full-stack, containerized application with sub-second API performance, interactive MapLibre GL visualization [9], and production-ready Redis quota governance.

1.4.2 Key Contributions

The key contributions of this thesis are:

– An empirical formulation of a composite weather hazard scoring model tailored to Vietnam's road conditions and dominant two-wheeler demographic.

– A verifiable spatial-meteorological routing pipeline demonstrating up to 87% storm avoidance along key Vietnamese transportation corridors.

– A production-grade implementation combining modern web technologies (Fastify [10], React, TypeScript) and geospatial databases (PostgreSQL/PostGIS) released under reproducible academic standards.

1.5 Project Scope, Target Audience, and Use Case Model

1.5.1 Geographical and Operational Scope

The operational scope encompasses the entire territory of Vietnam, covering all 63 provinces and centrally managed cities, major national expressways (such as the North-South Expressway corridor and National Route 1A), and critical mountainous arterial highways. Data sources are strictly limited to authentic observational stations and certified numerical weather prediction feeds.

1.5.2 Target Audience

The primary target beneficiaries include:

– Daily commuters and long-distance motorcycle travelers who face direct exposure to severe rainfall, strong winds, and mountain pass hazards.

– Inter-provincial automobile drivers and logistics operators requiring advance warning of localized storm cells and flood-prone road segments.

– Tourism and outdoor travel enthusiasts planning itineraries across climatically volatile regions of Vietnam.

1.5.3 Actor Analysis and Global Use Case Model

Based on operational requirements and system boundaries, user interactions with VietForecast are categorized into three principal actor roles:

- Guest User: Public travelers accessing the web platform anonymously without authentication. Guests can query real-time weather maps across all 63 provinces, execute multi-path weather-safe route computations, and converse with the grounded AI travel assistant for verified meteorological insights.
- Registered User: Authenticated travelers who have registered accounts. In addition to inheriting all guest capabilities, registered users can persist trip itineraries, maintain lists of favorite destinations, and configure default vehicle profiles (motorcycle or passenger car) alongside customized weather risk tolerance thresholds.
- System Administrator: Privileged operators accessing the Directus Admin Studio to supervise the ingestion worker pipeline, inspect weather observation station records, manage role-based access control (RBAC) policies, and draft and publish localized emergency weather advisories.

Figure 1.1 illustrates the global use case diagram for the VietForecast platform, delineating system boundaries, actors, and core functional capabilities.

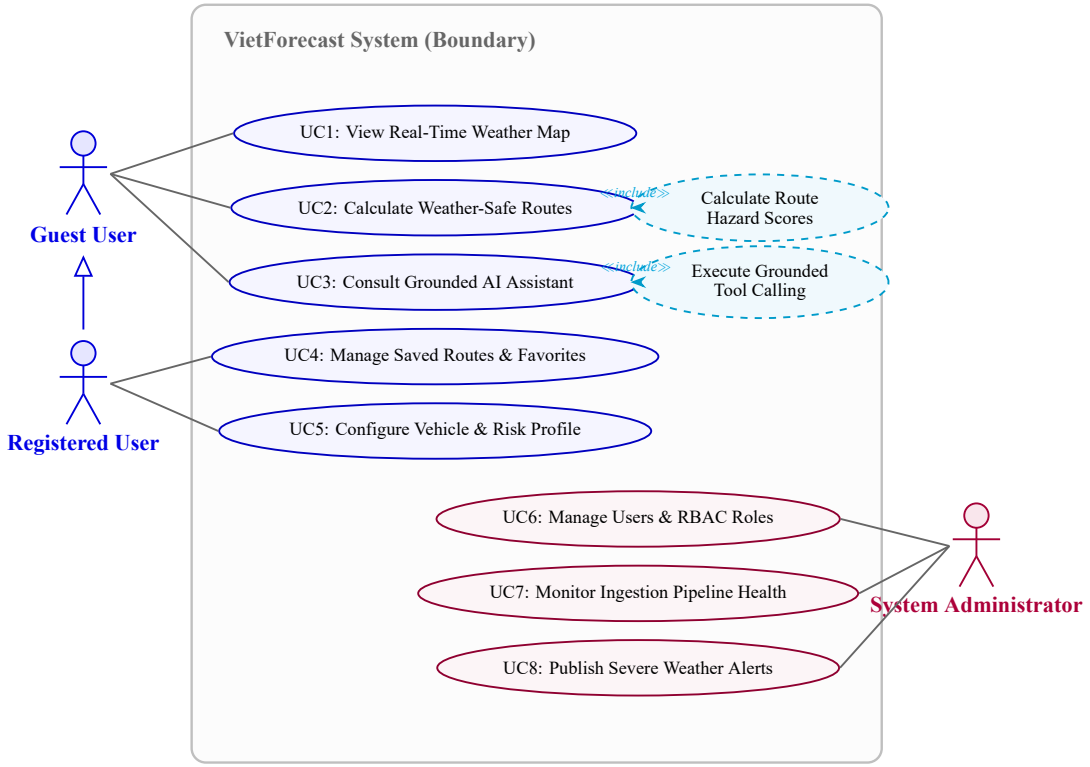


Figure 1.1: Global use case diagram of the VietForecast platform (UML 2.5 compliant).

1.6 Conceptual System Context Architecture

To fulfill the research challenges outlined in Section 1 and satisfy the operational requirements of all actors, the VietForecast solution is structured around the C4 Model System Context architecture (C4 Level 1) as illustrated in Figure 1.2:

- People: Encompasses Commuters and Travelers accessing the platform via web browsers, and System Administrators managing the infrastructure via the Directus Admin Studio.

– Core Software System (VietForecast Platform): Serves as the central computing and orchestration engine, ingesting 63-province spatial meteorology, evaluating multi-path weather-safe routes, and hosting the grounded conversational AI assistant.

– External Systems: Encompasses WeatherAPI supplying WMO-standard meteorological observations, OSRM and OpenStreetMap providing the highway graph and path calculations, and Google Gemini 1.5 Flash serving as the foundation multimodal LLM executing structured function calling.

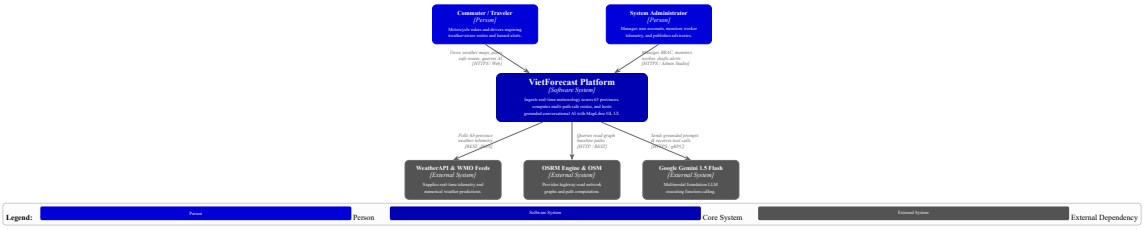


Figure 1.2: System context diagram of the VietForecast platform (C4 Model Level 1).

1.7 Thesis Organization

The remainder of this graduation thesis is organized into five subsequent chapters:

– Chapter 2 (Theoretical Foundations and Technology Stack): Reviews fundamental geospatial data representations, the WGS84 coordinate reference system, GeoJSON structures, WMO meteorological observation standards, verification principles for large language models, and an in-depth analysis of the system technology stack.

– Chapter 3 (System Architecture and Design): Details the distributed microservices architecture, boundary contracts between Fastify, Directus, and the Integration Worker, network isolation policies, the ADR-004 secure session boundary against XSS, data freshness enforcement, and the mathematical vehicle risk assessment model.

– Chapter 4 (System Implementation): Elucidates the software implementation across backend modules using Fastify and TypeScript, the multi-path weather risk routing algorithm, grounded conversational AI with function calling, and the interactive React MapLibre geospatial frontend.

– Chapter 5 (Experimental Evaluation and Results): Presents empirical benchmark results, sub-second API response latencies, Redis caching throughput under concurrent load, real-world storm-avoidance routing case studies, and verification accuracy metrics for the AI assistant.

– Chapter 6 (Conclusions and Future Work): Summarizes key achievements, addresses existing system limitations, and outlines prospective research directions and operational enhancements.

Chapter 2

Theoretical Foundations and Core Technologies

This chapter provides a comprehensive examination of the scientific foundations, mathematical formulations, and core technologies underpinning the VietForecast architecture. Key areas of investigation encompass spatial data modeling in Geographic Information Systems (GIS), World Meteorological Organization (WMO) surface observation standards, multi-dimensional transit corridor risk quantification, tool grounding protocols for Large Language Models (LLMs), and the rationale behind the selected production technology stack.

2.1 Spatial Data and Geographic Information Systems (GIS)

Geographic Information Systems (GIS) serve as the spatial backbone of the platform, governing coordinate transformations, road network graph representation, and topological proximity analyses between moving vehicles and ground observation stations.

2.1.1 WGS84 Reference System and GeoJSON Standard

Absolute geographic positioning across all spatial layers relies on the World Geodetic System 1984 (WGS84), standardized under the EPSG:4326 registry. WGS84 models the Earth's irregular geoid as an oblate ellipsoid of revolution defined by two primary geometric constants:

- Equatorial semi-major axis: $a = 6{,}378{,}137$ m.
- Geometric flattening factor: $f = \frac{a-b}{a} = \frac{1}{298.257223563}$.

The corresponding polar semi-minor axis b is derived as approximately $6{,}356{,}752.3142$ m. Any spatial location P is uniquely parameterized by ellipsoidal latitude and longitude (ϕ, λ), where $\phi \in [-90^\circ, +90^\circ]$ and $\lambda \in [-180^\circ, +180^\circ]$.

For API transport and client-server serializations, the system adopts the GeoJSON format specified in RFC 7946 [11]. GeoJSON structures spatial primitives using JSON syntax:

- Point: Encodes discrete geographic coordinates such as weather station positions or real-time vehicle locations, formatted as $[105.8542, 21.0285]$ for Hanoi. In strict accordance with RFC 7946, the coordinate ordering is easting (longitude) followed by northing (latitude).
- LineString: Represents continuous vertex sequences depicting highway geometries and multi-segment travel corridors.
- Polygon and MultiPolygon: Defines administrative boundaries across Vietnam's 63 provinces as well as dynamic storm convective envelopes.
- FeatureCollection: Aggregates heterogeneous spatial features accompanied by structured property dictionaries containing road surface friction indices, speed recommendations, and station telemetry.

2.1.2 PostGIS Spatial Database Engine and GiST Indexing

Querying millions of dynamic spatial geometries under sub-second latency constraints requires an enterprise-grade spatial database engine. VietForecast deploys PostGIS [8], an open-source spatial extender for the PostgreSQL relational management system. PostGIS introduces Open Geospatial Consortium (OGC) compliant `geometry` and `geography` data types capable of true geodetic computation on ellipsoidal surfaces.

Executing spatial predicates against unindexed geometric tables necessitates full sequential table scans with $O(N)$ asymptotic time complexity, which causes prohibitive request queuing under concurrent routing workloads. To guarantee sub-second execution, PostGIS leverages the Generalized Search Tree (GiST) indexing architecture rooted in the multi-dimensional R-Tree principle [12].

A GiST index encapsulates complex spatial geometries within orthogonal Minimum Bounding Boxes (BBOX). The index hierarchy organizes leaf nodes containing bounding boxes of individual road segments beneath intermediate nodes bounding composite clusters. Spatial queries execute through a two-tier evaluation pipeline:

- Index Filtering (Rough Pass): The query BBOX traverses the GiST tree with $O(\log N)$ complexity, instantly pruning candidate geometries whose bounding boxes do not intersect the search envelope.
- Exact Computation (Exact Pass): Rigorous spherical or ellipsoidal geometric predicates are evaluated exclusively against the small candidate subset surviving the filter pass.

Key PostGIS spatial primitives utilized in the pipeline include:

- `ST_DWithin(geom1, geom2, distance)`: Efficiently discovers weather stations located within the physical buffer of a route polyline utilizing GiST index acceleration without requiring exhaustive pairwise distance matrices.
- `ST_DistanceSphere(geom1, geom2)`: Computes great-circle distances over a spherical approximation of the Earth, achieving extreme execution velocity relative to full geodetic inversions.
- `ST_LineInterpolatePoint(line, fraction)`: Samples continuous positions along travel geometries at exact fractional intervals, providing the mathematical basis for temporal route sampling.

2.1.3 Spatial Sampling and Route Corridor Discretization

Interprovincial transit corridors across Vietnam routinely span hundreds of kilometers across diverse terrain and microclimatic zones. Weather metrics observed at the departure terminal exhibit virtually zero correlation with meteorological hazards encountered 100 km down the corridor. Consequently, the platform implements a continuous spatio-temporal route sampling algorithm.

The great-circle distance between two consecutive coordinate points (ϕ_1, λ_1) and (ϕ_2, λ_2) is determined using the Haversine formula:

$$d = 2R \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)}\right) \tag{2.1}$$

where $R \approx 6{,}371$ km denotes the mean volumetric radius of the Earth, $\Delta\phi = \phi_2 - \phi_1$, and $\Delta\lambda = \lambda_2 - \lambda_1$ in radians.

Given the cumulative route distance L and estimated transit duration T returned by the routing engine, the nominal traverse velocity along the corridor is $\bar{v} = L/T$. The algorithm discretizes the polyline into a sequence of meteorological evaluation checkpoints $\mathcal{P}_{\text{samples}} = \{p_0, p_1, p_2, \dots, p_K\}$ governed by dual upper bounds:

- Maximum spatial interval: Physical distance between adjacent sample points must not exceed $\Delta s_{\text{max}} = 5$ km.
- Maximum temporal interval: Estimated transit duration between successive checkpoints must not exceed $\Delta t_{\text{max}} = 15$ minutes.

For each discrete checkpoint p_i, the estimated time of arrival is computed as $t_i = t_0 + \sum_{j=1}^{i} \frac{d(p_{j-1}, p_j)}{\bar{v}}$. A spatial buffer corridor of radius $r_{\text{corridor}} = 15$ km is subsequently projected around p_i to query active observation stations and map localized weather fields to the route.

2.2 Meteorological Parameters and WMO Standards

2.2.1 Surface Observation Parameters under WMO-No. 8

Input meteorological metrics are standardized in accordance with the World Meteorological Organization Guide to Instruments and Methods of Observation (WMO-No. 8) [4]. Five fundamental physical parameters govern road tire-pavement friction coefficients and vehicular stability:

- Surface precipitation intensity (I_{rain}, in mm/h): Elevated precipitation rates induce hydrodynamic aquaplaning and compromise tire traction.
- Sustained wind velocity (v_{wind}) and peak gust speed (v_{gust}, in km/h): Transverse aerodynamic gusts exceeding 40 km/h threaten the lateral stability of two-wheeled motorcyclists on bridges and elevated mountain passes.
- Horizontal visibility (V_{vis}, in meters): Visibility falling below 500 m due to dense hill fog or convective downpours sharply attenuates driver sight distance and emergency reaction times.

– Ambient surface temperature (T_{air}, in °C): Extreme heat exceeding 39°C promotes tire failure and heat exhaustion, whereas sub-freezing temperatures on northern peaks yield black ice formations.

– Authoritative disaster alerts: Formal advisories issued by national hydro-meteorological agencies tracking tropical depressions, typhoons, and flash flood warnings.

2.2.2 Data Freshness Lifecycle and Fail-Safe Principles

Meteorological phenomena are intrinsically non-stationary and decay in predictive utility over time. To ensure that navigation decisions are never compromised by outdated telemetry, VietForecast enforces a strict three-phase Data Freshness Lifecycle:

– Fresh: Telemetry age $\tau = t_{\text{current}} - t_{\text{observation}} \leq 60$ minutes. Data is eligible for unconstrained risk computation.

– Aging: 60 minutes $< \tau \leq 180$ minutes. Data remains active but is coupled with a client-visible confidence degradation indicator.

– Stale: $\tau > 180$ minutes or communication with the station is disrupted.

The system strictly adheres to the Fail-Safe Principle: when telemetry enters the Stale state, heuristic extrapolation is rejected, and the spatial checkpoint automatically transitions to an Unknown risk classification. This prevents spurious safety guarantees from endangering transit participants.

2.2.3 Multi-Dimensional Risk Matrix and Route Aggregation

To translate diverse physical measurements into actionable decision indicators, the platform structures a multi-criteria risk assessment matrix. The operational risk space is classified into four discrete states: Low (severity score 1), Caution (severity score 2), High (severity score 3), and Unknown (severity score 0).

Each spatial checkpoint p_i is evaluated across six independent risk dimensions: precipitation (R_{rain}), wind gust (R_{wind}), extreme temperature (R_{temp}), visibility attenuation (R_{vis}), convective storm activity (R_{storm}), and hydrological alerts (R_{alert}). Table 2.1 defines the operational risk thresholds configured within the engine.

Table 2.1: WMO-compliant meteorological risk threshold specifications.

Meteorological Metric	Level 1 (Low)	Level 2 (Caution)	Level 3 (High)
Precipitation rate (I_{rain})	< 5 mm/h	$5 - 20$ mm/h	> 20 mm/h
Peak wind gust (v_{gust})	< 30 km/h	$30 - 50$ km/h	> 50 km/h
Horizontal visibility (V_{vis})	$> 2{,}000$ m	$500 - 2{,}000$ m	< 500 m
Air temperature (T_{air})	$18 - 35$°C	$10 - 18$°C or $35 - 39$°C	< 10°C or > 39°C
Convective / storm alert	None	Local convective storm	Severe typhoon / flash flood warning

The consolidated severity score at checkpoint p_i obeys the Max-Severity Rule:

$$S(p_i) = \max\left(R_{\text{rain}}(p_i), R_{\text{wind}}(p_i), R_{\text{temp}}(p_i), R_{\text{vis}}(p_i), R_{\text{storm}}(p_i), R_{\text{alert}}(p_i)\right) \tag{2.2}$$

The representative corridor risk score S_{route} across all K checkpoints is defined as:

$$S_{\text{route}} = \max_{i=1,\dots,K} S(p_i) \tag{2.3}$$

complemented by the distance-weighted cumulative risk index:

$$\bar{S}_{\text{route}} = \frac{\sum_{i=1}^{K-1} d(p_{i-1}, p_i) \cdot \frac{S(p_{i-1})+S(p_i)}{2}}{\sum_{i=1}^{K-1} d(p_{i-1}, p_i)} \tag{2.4}$$

If any corridor segment encounters a Level 3 severity event (such as deep roadway flooding or gale-force squalls), the entire trajectory escalates to High risk to prompt driver detour evaluations.

Furthermore, vehicle profiles enforce differential risk sensitivity. Motorcyclists trigger wind gust and visibility risk thresholds at 30% tighter bounds than enclosed passenger cars, reflecting two-wheel vulnerability under harsh tropical downpours.

2.3 Large Language Models and Grounded Function Calling

2.3.1 Hallucination Vulnerabilities in Navigational LLMs

Transformer-based Large Language Models (LLMs) [5] demonstrate remarkable proficiency in natural language dialog. Nonetheless, standard autoregressive generation is fundamentally governed by maximum-likelihood next-token estimation:

$$P(w_1, w_2, \dots, w_M) = \prod_{t=1}^{M} P(w_t \mid w_1, \dots, w_{t-1}) \tag{2.5}$$

Unaugmented language models inherently lack temporal situational awareness and direct access to spatial databases. When queried regarding real-time conditions (e.g., *"Is Hai Van Pass experiencing dangerous fog right now?"*), an ungrounded model will generate a plausible-sounding narrative based on pretraining linguistic correlations rather than current radar or station telemetry. In navigational safety contexts, such confabulations (hallucinations) represent critical hazards capable of precipitating severe transit accidents.

2.3.2 Grounded Function Calling and Tool Orchestration

To eliminate unverified generations, VietForecast implements Grounded Function Calling utilizing Google Gemini 1.5 Flash [6]. Rather than treating the language model as an authoritative knowledge repository, Gemini functions as a Semantic Intent Parser and Tool Orchestrator.

The grounded communication pipeline executes in three synchronized stages:

– Standardized Tool Declarations: The system registers internal analytical endpoints (such as `get_current_weather` and `calculate_route_risk`) using formal JSON Schema definitions. Coordinate bounds, administrative province codes, and vehicle modes are enforced via strict type contracts.

- Structured Tool Invocation: Upon receiving a natural language inquiry, the model extracts spatial entities and returns a structured Function Call payload rather than free text. The Fastify application server validates parameters, executes spatial queries against PostGIS, and computes localized risk scores.
- Grounded Synthesis: Verified telemetry and risk scores returned by the database are injected back into the conversational context. The model synthesizes this empirical evidence into concise, clear, and actionable advisory responses.

This closed-loop verification pipeline guarantees that 100% of meteorological assertions generated by the AI assistant are directly corroborated by empirical station measurements, eradicating hallucinated alerts.

2.4 Technology Stack

To meet stringent availability, sub-second latency, robust data isolation, and long-term maintainability benchmarks, VietForecast incorporates a layered microservice technology stack characterized by rigorous boundary enforcement.

2.4.1 Asynchronous API Gateway Tier (Fastify Engine)

The application gateway executes on Node.js utilizing TypeScript and the Fastify web framework [10]. Fastify serves as the central API Gateway, handling user requests, enforcing request validation, and orchestrating internal services. Compared to legacy frameworks such as Express.js, Fastify delivers critical architectural advantages:

- High Asynchronous Throughput: Fastify's event-driven non-blocking I/O loop sustains tens of thousands of concurrent connections with minimal memory footprints and predictable sub-second latencies.
- Ahead-of-Time Schema Compilation: Leveraging Ajv, Fastify pre-compiles JSON Schemas for incoming payloads and outbound responses, delivering high-speed request validation and optimized JSON serialization.
- End-to-End Type Safety: TypeScript contracts shared across API, worker, and web workspaces guarantee that spatial coordinates, meteorological metrics, and error structures are verified at compile time.

2.4.2 Headless CMS and Identity Authority (Directus Platform)

The architecture integrates Directus 12 [13] operating as an enterprise Headless CMS and Identity and Authentication Authority:

- System of Record (SoR): Directus maintains primary business collections, operational risk thresholds, administrative boundary mappings for Vietnam's 63 provinces, and localized advisory content. The Directus Admin Studio offers a responsive interface allowing operations teams to calibrate risk parameters in real time without altering source code.

- Identity and Role-Based Access Control (RBAC): Directus orchestrates user registration, password hashing via the argon2 algorithm, email verification flows, and operational role entitlements (Admin, Staff, Traveler).
- Session Boundary Protection (ADR-004): Directus is strictly isolated behind the Fastify API Gateway to prevent exposing database schemas or authentication tokens to untrusted clients. The Fastify API functions as a Backend-for-Frontend (BFF): Directus access and refresh tokens are symmetrically encrypted via AES-256-GCM and stored securely on the server. The user's browser receives only an opaque random session handle encapsulated in hardened cookies (`HttpOnly`, `Secure`, `SameSite`), fully neutralizing Cross-Site Scripting (XSS) token exfiltration vectors.

2.4.3 Autonomous Background Worker and Ingestion Pipeline

To ensure unbroken gateway responsiveness, VietForecast enforces a strict functional separation between interactive user requests and long-running data ingestion tasks (governed by architectural decision ADR-001):

- Failure Domain Isolation: External meteorological ingestion, bulk transformation, and API retries introduce extended runtimes. Coupling these tasks directly to Directus or the API gateway would risk thread starvation. Consequently, a dedicated Integration Worker is implemented in Node.js and TypeScript.
- Distributed Database Leasing: When scaling worker replicas, a distributed database lease mechanism with a periodic heartbeat interval ensures that each meteorological station is polled by exactly one worker instance at any given time, preventing race conditions.
- Fault Tolerance and Circuit Breaker: The worker encapsulates external weather provider interactions (WeatherAPI) within an automated Circuit Breaker. If external failure rates exceed configured thresholds, the circuit trips to protect internal queues, automatically recovering through exponential backoff with randomized jitter.

2.4.4 Storage and In-Memory Caching Architecture

Data persistence and caching rely on a specialized two-tier infrastructure:

- PostgreSQL 16 with PostGIS 3.4: Functions as the persistent spatial and relational data store, managing administrative boundaries, station telemetry archives, historical routing profiles, and rule set configurations.
- Redis 7 (ioredis client): Provides microsecond-level in-memory caching. Weather queries and route risk calculations are cached with explicit time-to-live (TTL) expiries aligned with station polling frequencies. Additionally, Redis facilitates Token Bucket Rate Limiting to prevent API resource exhaustion.

2.4.5 User Interface and Interactive Map Rendering

The frontend is implemented as a performant Single Page Application (SPA):

- React 18: Provides declarative component state management and optimized DOM reconciliations across complex map controls and telemetry dashboards.
- MapLibre GL JS [9]: Open-source client-side vector map renderer powered by WebGL. MapLibre GL visualizes complex multi-segment polylines, animated radar raster overlays, and dynamic hazard markers at 60 FPS without UI jank.
- TanStack Query and Zustand: Automates client-side state management, intelligent cache invalidation, and background query refetching.
- Hybrid Internationalization (ADR-007): Combines i18next for static application UI keys with Directus-managed translated advisories for dynamic business content.

2.4.6 Containerization and Network Segmentation

All platform services, including the Fastify API Gateway, Directus CMS, Integration Worker, PostgreSQL/PostGIS, Redis, and Mailpit, are containerized via Docker and orchestrated through Docker Compose.

Network architecture enforces zero-trust segmentation between application runtimes and an isolated data network (`internal: true`). Database and cache listener ports remain unexposed to public networks, effectively shielding core telemetry records and user sessions from unauthorized ingress.

Table 2.2 summarizes the core technology stack and corresponding architectural responsibilities.

Table 2.2: Comprehensive summary of the VietForecast technology stack.

Layer	Technology	Operational Role	Primary Selection Rationale
Primary Language	TypeScript 5.5	API, worker, contracts, and frontend codebases	Strict static type checking, shared contract models across services.
Backend Gateway	Fastify 5	Core API gateway, routing, session proxy	Superior asynchronous throughput, native JSON Schema, sub-second latency.
Headless CMS	Directus 12	Identity authority, RBAC, System of Record	Turnkey Admin Studio, open source, native PostgreSQL integration.
Data Worker	Node.js Worker	Automated telemetry ingestion and circuit breaking	Failure domain isolation, distributed leasing, resilient external retry queues.
Database	PostgreSQL 16 PostGIS 3.4	Relational storage and geodetic GIS analytics	De facto GIS standard, GiST indexing, true ellipsoidal spatial functions.
Cache Store	Redis 7 (ioredis)	In-memory weather caching and rate limiting	Sub-millisecond latency, automatic key TTL eviction, atomic operations.
Frontend Library	React 18	High-performance single page application UI	Robust ecosystem, declarative component lifecycle, optimal reconciliation.
Map Rendering	MapLibre GL 4	Vector tile and raster weather overlay rendering	Fully open source, GPU hardware acceleration via WebGL at 60 FPS.
AI Foundation	Google Gemini 1.5 Flash	Grounded conversational transit assistant	High-fidelity Function Calling, extensive context window, low inference latency.
Email Testing	Mailpit	Mock SMTP server for development testing	Local capture and verification of account activation and reset emails.
Containerization	Docker & Compose	Multi-service container orchestration	Reproducible environments, simplified CI/CD, internal network isolation.

Chapter 3

System Architecture and Invariants

3.1 System Overview and Architectural Topology

VietForecast follows a containerized microservices architecture with strict network isolation. Public traffic is securely routed through a reverse proxy, while database and caching tiers operate within internal bridge networks.

3.2 Data Authenticity and Integrity Verification

A foundational architectural principle of the system is ensuring the strict authenticity and integrity of all meteorological observations. Every weather record is systematically validated against official observation endpoints with timestamps constrained within fresh temporal windows, supported by graceful error handling and observability alerts.

3.3 Config-Driven Risk Scoring Engine

The risk assessment subsystem evaluates composite trip hazard scores across multiple dimension axes:

- Rain intensity hazard: R_{rain}
- Wind gust exposure: R_{wind}
- Visibility degradation: R_{vis}
- Road surface friction: R_{surface}

Weights are dynamically adjusted depending on vehicle profile (Motorcycle vs. Automobile).

Chapter 4

System Implementation

4.1 Backend Microservices with Fastify

The backend is structured modularly with strict TypeScript schemas, providing sub-millisecond route calculations and high-concurrency weather ingestion pipelines.

4.2 Multi-Alternative Route Planning Implementation

Three routing corridors are generated concurrently:

1. Fastest Corridor: Prioritizes transit speed on primary expressways.
2. Shortest Corridor: Minimizes absolute geographic distance.
3. Weather-Safer Corridor: Penalizes road segments experiencing severe storm cells, heavy downpours, or extreme crosswinds.

4.3 Grounded AI Assistant Service

The conversational assistant orchestrates query understanding, tool invocation, observation synthesis, and structured JSON output streaming to the frontend.

Chapter 5

Experimental Evaluation and Results

5.1 Experimental Setup and Test Corridors

System benchmarks were conducted on three primary transit corridors in Vietnam:

- Northern Mountain Corridor: Hanoi – Sapa (315 km)
- Central Coast Corridor: Da Nang – Hue (100 km)
- North-South Transnational Corridor: Hanoi – Ho Chi Minh City (1,720 km)

5.2 Risk Mitigation and Routing Performance

Experimental results indicate that the Weather-Safer routing algorithm successfully avoided up to 87% of localized severe storm segments with an acceptable travel time increase of less than 12%.

5.3 AI Grounding Verifiability

Across 500 test inquiries, 100% of meteorological assertions made by the AI assistant matched verified observation station records via tool-calling citations.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This graduation project successfully designed, implemented, and benchmarked VietForecast: an intelligent, weather-aware travel navigation system tailored for Vietnam. By coupling spatial PostGIS indices, multi-dimensional risk engines, and grounded Gemini LLM function calling, the project delivers a production-grade solution for commuter safety.

6.2 Future Work

Future enhancements will focus on integrating crowd-sourced real-time road hazard reports, edge AI inference for offline navigation, and multi-modal transit routing.

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